# Automated Chest X-ray Report Generation Remains Unsolved


Xiaoman Zhang [1], Julián N. Acosta [1], Xiaoli Yang [1], Subathra Adithan [1], Luyang Luo [1], Hong-Yu Zhou [1], Joshua Miller [2], Ouwen Huang [2,3,4], Zongwei Zhou [5], Ibrahim Ethem Hamamci [6], Shruthi Bannur [7], Kenza Bouzid [7], Xi Zhang [8], Zaiqiao Meng [8], Aaron Nicolson [9], Bevan Koopman [9], Inhyeok Baek [10], Hanbin Ko [11], Mercy Prasanna Ranjit [12], Shaury Srivastav [12], Sriram Gnana Sambanthan [13], Pranav Rajpurkar[1]

[1] Department of Biomedical Informatics, Harvard Medical School, USA
[2] Gradient Health, Durham, NC, USA [3] Department of Statistical Science, Duke University, Durham, NC, USA [4] Laplace Institute, Durham, NC, USA
[5] Department of Computer Science, Johns Hopkins University, MD, USA [6] University of Zurich, Switzerland [7] Microsoft Research Health Futures, USA
[8] University of Glasgow, UK [9] Australian e-Health Research Centre, CSIRO Health and Biosecurity, Brisbane, Australia [10] Seoul National University, South Korea
[11] Seoul National University Graduate School, South Korea [12] Microsoft Research India, India [13] Indian Institute of Technology Madras, Chennai, India


## Motivation

1. Automated radiology report generation shows promise but lacks standardized evaluation frameworks, preventing reliable comparative analysis across different models.
2. Existing public datasets serve dual purposes as both training and evaluation data, failing to adequately test models' ability to generalize to new, unseen clinical distributions.
3. Current AI systems have not yet achieved the reliability required for widespread clinical applications, but the limitation remains unclear without comprehensive benchmarking.

## Contributions

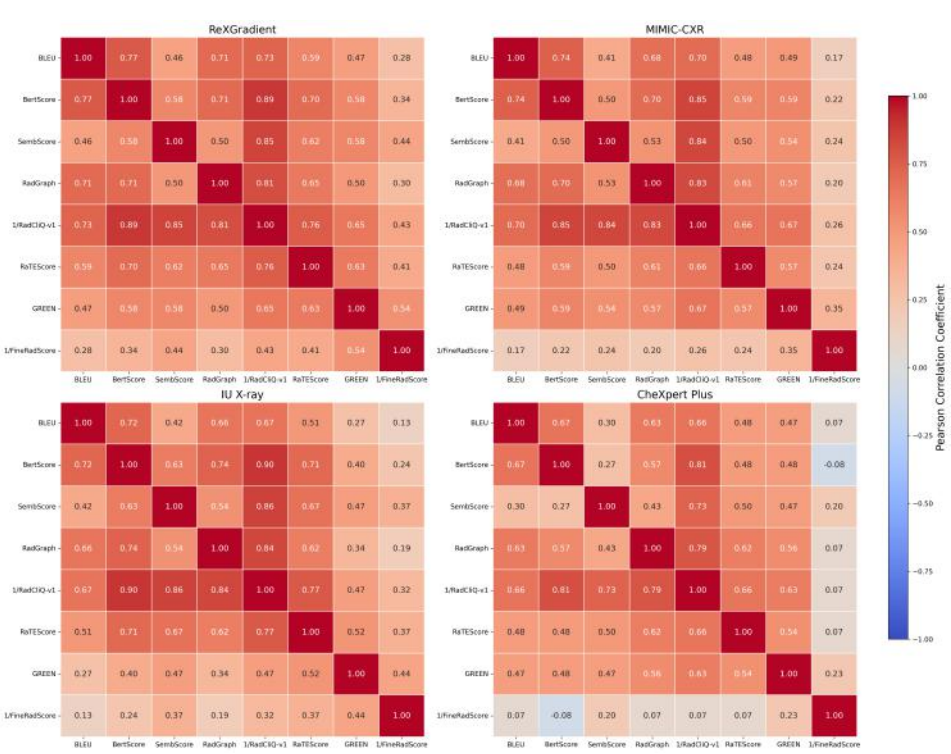
1. We conducted the ReXrank Challenge V1.0, evaluating **24** models (**8 new submissions** + 16 previously benchmarked) on the largest and most diverse test dataset available.
2. We performed comprehensive analysis using **8** distinct evaluation metrics across **10,000** studies from **67** U.S. medical institutions through the ReXGradient dataset.
3. We provide evidence that automated chest X-ray report generation remains fundamentally unsolved, with top models achieving less than **45%** error-free reporting on abnormal cases.

## Results

**ReXGradient Results.** No single model dominates across all metrics, with substantial variation in rankings depending on evaluation approach

| Model-Name | BLEU | BertScore | SembScore | RadGraph | 1/RadCliQ-V1 | RATEScore | GREEN | 1/FineRadScore |
|---|---|---|---|---|---|---|---|---|
| MedVersa | 0.210 | 0.431 | **0.498** | 0.202 | 1.008 | 0.527 | 0.532 | **0.475** |
| MAIRA-2 | 0.205 | **0.436** | 0.462 | 0.187 | 0.963 | 0.559 | 0.531 | **0.475** |
| Libra | 0.176 | 0.408 | 0.474 | 0.169 | 0.913 | 0.544 | 0.549 | 0.473 |
| CXRMate-ED | 0.202 | 0.398 | 0.415 | 0.187 | 0.872 | 0.564 | 0.518 | 0.472 |
| CheXpertPlus-MIMIC | 0.154 | 0.341 | 0.440 | 0.131 | 0.778 | 0.501 | 0.517 | 0.471 |
| MoERad-IU | **0.227** | 0.434 | 0.446 | **0.247** | **1.018** | **0.575** | 0.494 | 0.468 |
| CvT2DistilGPT2-MIMIC | 0.186 | 0.374 | 0.458 | 0.175 | 0.866 | 0.522 | 0.510 | 0.468 |
| CXRMate | 0.169 | 0.363 | 0.479 | 0.174 | 0.863 | 0.545 | **0.550** | 0.466 |
| CheXpertPlus-CheX-MIMIC | 0.169 | 0.372 | 0.440 | 0.153 | 0.828 | 0.516 | 0.486 | 0.463 |
| DD-LLaVA-X | 0.166 | 0.387 | 0.469 | 0.174 | 0.886 | 0.542 | 0.504 | 0.459 |
| CXRMate-RRG24 | 0.150 | 0.327 | 0.462 | 0.152 | 0.792 | 0.518 | 0.408 | 0.458 |
| RadPhi3.5Vision | 0.209 | 0.383 | 0.488 | 0.169 | 0.891 | 0.544 | 0.453 | 0.458 |
| RGRG | 0.190 | 0.391 | 0.470 | 0.169 | 0.888 | 0.540 | 0.487 | 0.458 |
| CvT2DistilGPT2-IU | 0.176 | 0.394 | 0.405 | 0.166 | 0.839 | 0.518 | 0.467 | 0.456 |
| CheXagent | 0.093 | 0.305 | 0.366 | 0.080 | 0.674 | 0.428 | 0.241 | 0.455 |
| RaDialog | 0.188 | 0.402 | 0.450 | 0.158 | 0.876 | 0.522 | 0.435 | 0.454 |
| VLCI-MIMIC | 0.158 | 0.310 | 0.400 | 0.122 | 0.720 | 0.487 | 0.473 | 0.453 |
| VLCI-IU | 0.214 | 0.365 | 0.466 | 0.213 | 0.804 | 0.571 | 0.532 | 0.451 |
| BioMedGPT-IU | 0.099 | 0.317 | 0.437 | 0.157 | 0.771 | 0.472 | 0.388 | 0.450 |
| MoERad-MIMIC | 0.145 | 0.351 | 0.406 | 0.116 | 0.756 | 0.508 | 0.431 | 0.446 |
| RadFM | 0.157 | 0.365 | 0.392 | 0.135 | 0.775 | 0.504 | 0.406 | 0.437 |
| GPT4V | 0.075 | 0.214 | 0.337 | 0.138 | 0.629 | 0.470 | 0.497 | 0.428 |
| CheXpertPlus-CheX | 0.144 | 0.361 | 0.428 | 0.124 | 0.785 | 0.475 | 0.407 | 0.414 |
| LLM-CXR | 0.043 | 0.182 | 0.142 | 0.029 | 0.507 | 0.317 | 0.044 | 0.326 |

**Model Correlations.** Traditional text-matching metrics correlate poorly with LLM-based clinical metrics (correlations <0.60), indicating they capture different aspects of report quality and clinical utility.




## Challenge Design

**Duration:** December 1, 2024 - March 15, 2025
**Participants:** 8 new model submissions from 5 institutions + 16 previously benchmarked models
**Task:** Generate "Findings" section of radiology reports
**Evaluation Framework:**
Traditional (BLEU, BERTScore), Clinical (SembScore, RadGraph, RadCliQ, RaTEScore), LLM-based (GREEN, FineRadScore)
**Total Models Evaluated:** 24 models from diverse institutions

**Public Datasets**

MIMIC-CXR | IU X-ray | CheXpert Plus

**Private Datasets**

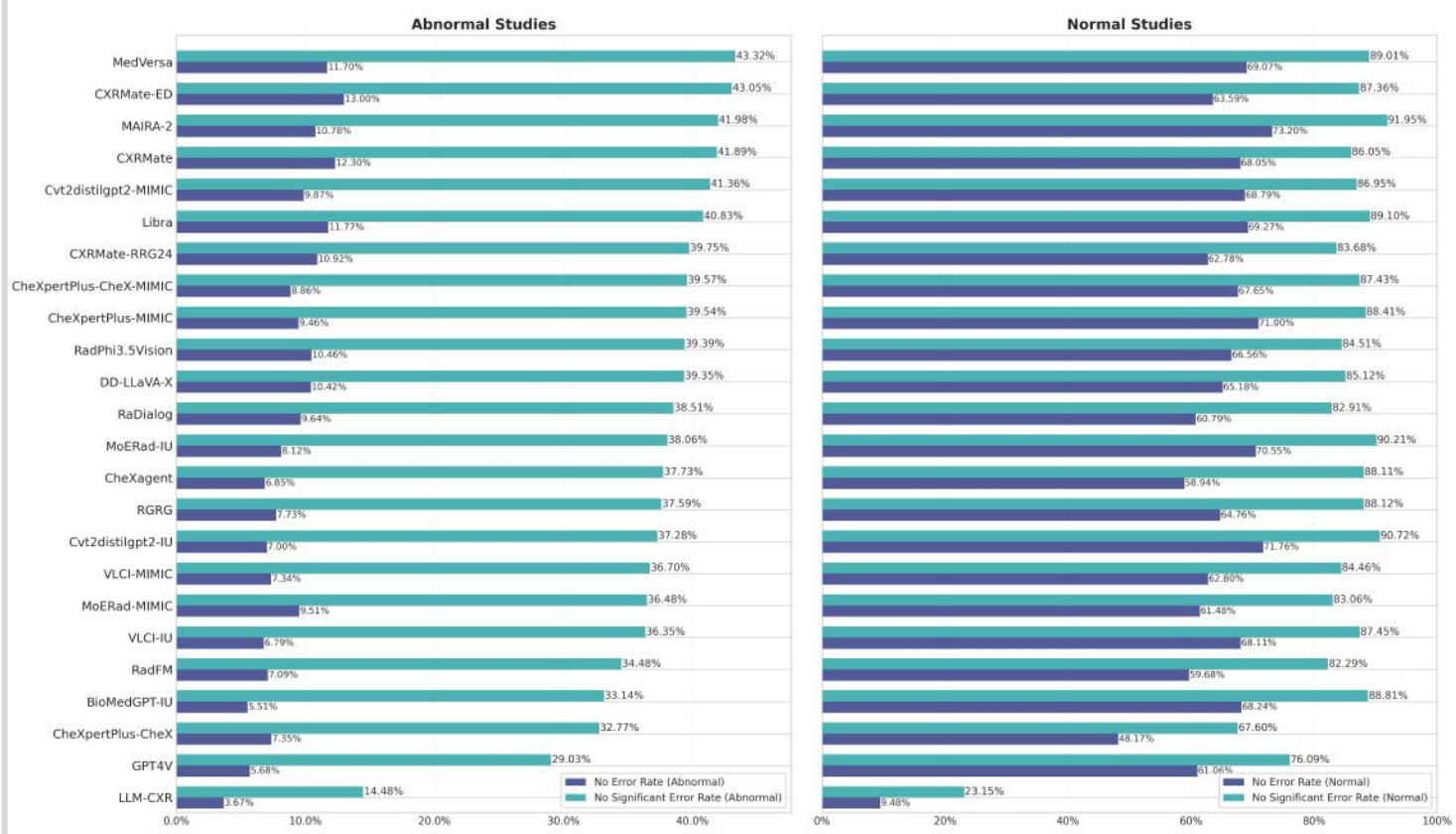
ReXGradient

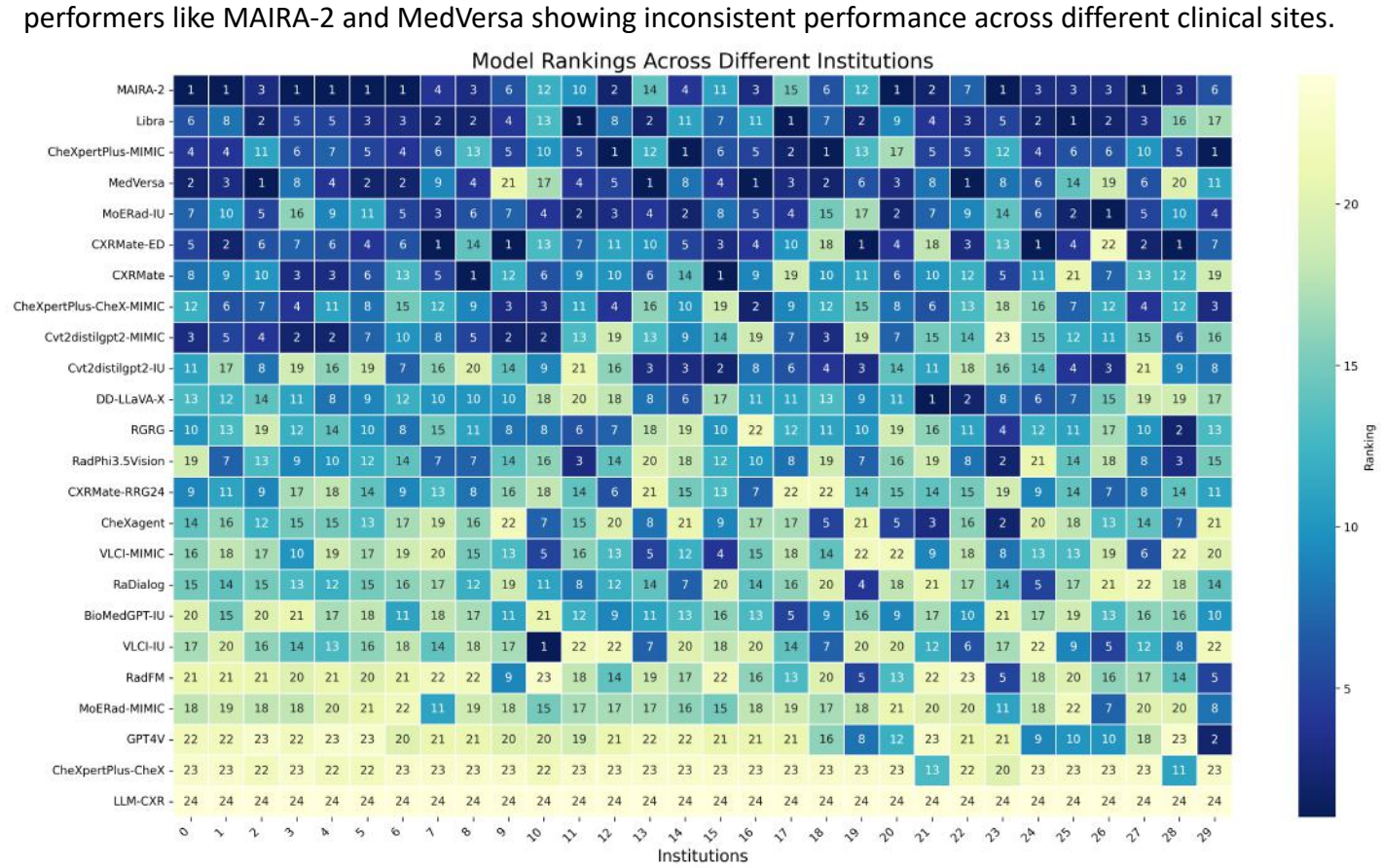
**Input Sample**

**Age**: 70-80. **Gender**: F.
**Indication**: Nausea, vomiting and abdominal pain. Question free air.
**Comparison**: None.

Frontal Image | Lateral Image

**Error Rates - Normal vs Abnormal.** AI models achieve 60-75% no actionable error rates on normal studies but only 5-15% on abnormal cases, revealing a critical performance gap where clinical accuracy matters most.




**Site Performance Variability.** Model rankings vary dramatically across healthcare institutions, with even top performers like MAIRA-2 and MedVersa showing inconsistent performance across different clinical sites.